

Project : Divide and conquer posterior sampling for Denoising Diffusion Priors

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1 Introduction

Deep Generative Models, and specifically Denoising Diffusion Probabilistic Models (DDPM), have emerged as powerful priors for solving inverse problems. Given an observation y related to a clean signal X_0 through a likelihood $g_0(y|x_0)$, the goal is to sample from the posterior distribution $p(X_0|y)$. While Diffusion Posterior Sampling (DPS) provides a tractable framework by approximating the posterior mean at each timestep, it often suffers from high approximation errors in the early stages of the reverse process (high k), where the Gaussian assumption of the denoised estimate is poor.

In this project, we explore the **Divide and Conquer Posterior Sampling (DCPS)** algorithm. The core idea is to decompose the global posterior sampling problem into several intermediate sub-problems. By leveraging the fact that intermediate distributions (the "bridge" between two close timesteps) are more Gaussian than the long-term projection to X_0 , DCPS reduces the accumulation of errors.

In this report, we

- Proved the assumptions made in the original paper to build the DDPM prior and the generative model.
- Provided an exact calculation of the bound of the wasserstein distance between the approximation and the true intermediate distribution for different examples. Those calculations confirm the bound given in Theorem 4.1, which is the main theoretical result motivating the DCPS algorithm.
- Explicitly formulate the two steps that represent the core of the DCPS algorithm
- Provided the explicit computation of each step of the algorithm in the case of a linear Gaussian likelihood and in particular for Gaussian mixture models.
- Implemented the DCPS algorithm for gaussian mixtures and sample from the posterior for four different likelihoods

We also join at the end some technical results on gaussians that are used throughout the report.

2 DDM priors

2.1 Noising the data

Suppose we can access sample from some data distribution p_{data} on $\mathbb{R}^d$. Let $X_0 \sim p_{\text{data}}$ and $\epsilon_k \sim \mathcal{N}(0, I_d)$ iid for $k \geq 1$.

Let α_k in $]0, 1]$ with $\alpha_0 = 1$ be a decreasing sequence close to zero. Let X_k an auto-regressive process defined by :

$$X_{k+1} = \sqrt{\frac{\alpha_{k+1}}{\alpha_k}} X_k + \sqrt{1 - \frac{\alpha_{k+1}}{\alpha_k}} \epsilon_{k+1}, \quad k \geq 0.$$

Assumption 2.1. (Forward process) Let $q_{k|0}(\cdot|x_0)$ be the distribution of X_k conditionally to $X_0 = x_0$. Let q_k be the distribution of X_k . Then, for all $k \geq 0$, we have

$$X_k = \sqrt{\alpha_k} X_0 + \sqrt{1 - \alpha_k} \xi_k,$$

where $\xi_k \sim \mathcal{N}(0, I_d)$ are iid and $X_0 \sim p_{\text{data}}$ are independent. As a consequence,

$$q_{k|0}(\cdot|x_0) \sim \mathcal{N}(\sqrt{\alpha_k} x_0, (1 - \alpha_k) I_d).$$

In particular, for all $k \geq 0$ and $x_k \in \mathbb{R}^d$, we have

$$q_k(x_k) = \int p_{\text{data}}(x_0) q_{k|0}(x_k|x_0) dx_0.$$

Proof. By induction on k . For $k = 0$, the result is immediate since $\alpha_0 = 1$. Suppose the result holds for some $k \geq 0$. Then,

$$\begin{aligned} X_{k+1} &= \sqrt{\frac{\alpha_{k+1}}{\alpha_k}} X_k + \sqrt{1 - \frac{\alpha_{k+1}}{\alpha_k}} \epsilon_{k+1} \\ &= \sqrt{\frac{\alpha_{k+1}}{\alpha_k}} (\sqrt{\alpha_k} X_0 + \sqrt{1 - \alpha_k} \epsilon_k) + \sqrt{1 - \frac{\alpha_{k+1}}{\alpha_k}} \epsilon_{k+1} \\ &= \sqrt{\alpha_{k+1}} X_0 + \sqrt{\alpha_{k+1} \frac{1 - \alpha_k}{\alpha_k}} \epsilon_k + \sqrt{1 - \frac{\alpha_{k+1}}{\alpha_k}} \epsilon_{k+1}. \end{aligned}$$

Let β_k the sum of the two last terms. Since ϵ_k and ϵ_{k+1} are independent standard Gaussian variables, β_k is a Gaussian variable with zero mean and variance $\alpha_{k+1} \frac{1 - \alpha_k}{\alpha_k} + 1 - \frac{\alpha_{k+1}}{\alpha_k} = 1 - \alpha_{k+1}$. This concludes the induction. $\square$

2.2 Learning the noise

The X_k corresponds to data after several steps of noise. They follow the distribution q_k . Thanks to those "new data", we want to learn how to approximate the noise ξ_k that was added to X_0 to get X_k . Let $\hat{\xi}_k^\theta$ be a neural network with parameters θ that takes as input X_k and outputs an approximation of ξ_k . DDMs learn to minimize the denoising objective :

$$\sum_{k=1}^T w_k \mathbb{E}_{\xi_k} \left[\|\hat{\xi}_k^\theta(\sqrt{\alpha_k} X_0 + \sqrt{1 - \alpha_k} \xi_k) - \xi_k\|^2 \right] \quad (1)$$

where the expectation is taken over $X_0 \sim p_{\text{data}}$ and $\xi_k \sim \mathcal{N}(0, I_d)$

Assumption 2.2. (*Optimal denoising*) Let $\mathcal{X} = \mathbb{R}^d$. Let

$$(\hat{\xi}_k^*)_{k=1}^T \in \arg \min_{(\hat{\xi}_k)_{k=1}^T \in (\mathcal{X})^T} \sum_{k=1}^T w_k \mathbb{E}_{X_0, Z} [L_k(\hat{\xi}_k, X_0 Z)]$$

where $L_k(f, X_0, Z) = \|f(\sqrt{\alpha_k}X_0 + \sqrt{1 - \alpha_k}Z) - Z\|^2$. Then, for all $k \geq 0$ and $x_k \in \mathbb{R}^d$, we have

$$\hat{\xi}_k^*(x_k) = \frac{x_k - \sqrt{\alpha_k} \mathbb{E}[X_0 | X_k = x_k]}{\sqrt{1 - \alpha_k}} = \mathbb{E}(\xi_k | X_k = x_k). \quad (2)$$

Proof. Because each term of the sum is positive, each $\hat{\xi}_k^{\theta^*}$ is the minimizer of

$$\begin{aligned} & \mathbb{E} \left[\|\hat{\xi}_k^{\theta^*}(\sqrt{\alpha_k}X_0 + \sqrt{1 - \alpha_k}\xi_k) - \xi_k\|^2 \right] \\ &= \mathbb{E} \left[\left\| \hat{\xi}_k^{\theta^*}(X_k) - \frac{X_k - \sqrt{\alpha_k}X_0}{\sqrt{1 - \alpha_k}} \right\|^2 \right] \\ &= \mathbb{E} \left[\mathbb{E} \left[\left\| \hat{\xi}_k^{\theta^*}(X_k) - \frac{X_k - \sqrt{\alpha_k}X_0}{\sqrt{1 - \alpha_k}} \right\|^2 \middle| X_k \right] \right] \end{aligned}$$

Everything in this expectation is positive, which means that for all $x_k \in \mathbb{R}^d, k \geq 0$, $\hat{\xi}_k^{\theta^*}(x_k)$ is the minimizer of the function $\xi \mapsto \mathbb{E} \left[\left\| \xi - \frac{x_k - \sqrt{\alpha_k}X_0}{\sqrt{1 - \alpha_k}} \right\|^2 \middle| X_k = x_k \right]$. By finding the zero of its gradient, it is classical that the minimum of this function is attained for

$$\begin{aligned} \hat{\xi}_k^{\theta^*}(x_k) &= \mathbb{E} \left[\frac{x_k - \sqrt{\alpha_k}X_0}{\sqrt{1 - \alpha_k}} \middle| X_k = x_k \right] \\ &= \frac{x_k - \sqrt{\alpha_k} \mathbb{E}[X_0 | X_k = x_k]}{\sqrt{1 - \alpha_k}}. \end{aligned}$$

□

This shows that $\hat{\xi}_k^{\theta^*}$ can be used to approximate the posterior mean of ξ_k given X_k . We can use this to approximate other quantities of interest.

Assumption 2.3. (*Posterior mean and score function approximation*)

Let $\hat{x}_{0|k}^*$ and $\hat{s}_k^*$ be functions defined by

$$\hat{x}_{0|k}^*(x_k) := \frac{1}{\sqrt{\alpha_k}} \left(x_k - \sqrt{1 - \alpha_k} \hat{\xi}_k^*(x_k) \right)$$

$$\hat{s}_k^*(x_k) := -\frac{x_k - \sqrt{\alpha_k} \hat{x}_{0|k}^*(x_k)}{1 - \alpha_k}.$$

Then :

$$\hat{x}_{0|k}^*(x_k) = \mathbb{E}[X_0 | X_k = x_k]. \quad (3)$$

$$\hat{s}_k^*(x_k) = \nabla_{x_k} \log q_k(x_k). \quad (4)$$

Proof. The first point is a straightforward consequence of Equation (2). The second point is a consequence of Tweedie's formula (Lemme 6.1), with $Z = X_k/\sqrt{\alpha_k}$ and $X = X_0$, which states that

$$\mathbb{E}[X_0|X_k = x_k] = \frac{x_k}{\sqrt{\alpha_k}} + \frac{1 - \alpha_k}{\sqrt{\alpha_k}} \nabla_{x_k} \log q_k(x_k).$$

Reinjecting the expression we proved on the first point, we have

$$\begin{aligned} \nabla_{x_k} \log q_k(x_k) &= \frac{\sqrt{\alpha_k} \hat{x}_0^*(x_k) - x_k}{1 - \alpha_k} \\ &= \hat{s}_k^*(x_k). \end{aligned}$$

□

2.3 Use of learned noise to build backward transitions

Let θ^* be the optimal parameters minimizing the denoising objective. Let $(t_k)_{k=0}^n$ be an increasing sequence in $0, \dots, T$ of integers with $t_0 = 0$. We suppose t_n is large enough so that q_{t_n} is close to a Gaussian distribution. For convenience, the paper denotes q_{t_k} by q_k . Then the paper makes the following assumption.

Assumption 2.4. (*Bridge Kernel*) Let $q_{j|0,k}(\cdot|x_0, x_k)$ be a bridge kernel. Then for all $0 \leq j < k \leq n$ and $x_0, x_k \in \mathbb{R}^d$, we have

$$q_{j|0,k}(\cdot|x_0, x_k) \sim \mathcal{N}(\mu_{j|0,k}(x_0, x_k), \sigma_{j|k}^2 I_d),$$

where

$$\begin{aligned} \mu_{j|0,k}(x_0, x_k) &= \frac{\sqrt{\alpha_j}(1 - \alpha_k/\alpha_j)}{1 - \alpha_k} x_0 + \frac{\sqrt{\alpha_k/\alpha_j}(1 - \alpha_j)}{1 - \alpha_k} x_k, \\ \sigma_{j|k}^2 &= \frac{(1 - \alpha_j)(1 - \alpha_k/\alpha_j)}{1 - \alpha_k} \end{aligned}$$

. In the following, we will often note $\lambda_{j,k} := \frac{\sqrt{\alpha_j}(1 - \alpha_k/\alpha_j)}{1 - \alpha_k}$ and $\beta_{j,k} := \frac{\sqrt{\alpha_k/\alpha_j}(1 - \alpha_j)}{1 - \alpha_k}$.

Proof. Bayesian rules gives for all $x_j \in \mathbb{R}^d$,

$$q_{j|0,k}(x_j|x_0, x_k) \propto q_{k|j}(x_k|x_j)q_{j|0}(x_j|x_0).$$

By Assumption 1,

$$\begin{aligned} q_{k|j}(\cdot|x_j) &\sim \mathcal{N}\left(\sqrt{\frac{\alpha_k}{\alpha_j}}x_j, \left(1 - \frac{\alpha_k}{\alpha_j}\right)I_d\right) \\ q_{j|0}(\cdot|x_0) &\sim \mathcal{N}(\sqrt{\alpha_j}x_0, (1 - \alpha_j)I_d), \end{aligned}$$

thus by reinjecting those expressions into the previous equation, we have

$$\begin{aligned}
q_{j|0,k}(x_j|x_0, x_k) &\propto \exp\left(-\frac{\|x_k - \sqrt{\frac{\alpha_k}{\alpha_j}}x_j\|^2}{2(1 - \frac{\alpha_k}{\alpha_j})}\right) \exp\left(-\frac{\|x_j - \sqrt{\alpha_j}x_0\|^2}{2(1 - \alpha_j)}\right) \\
&\propto \exp\left(-\frac{1}{2}x_j^T x_j \frac{(\frac{\alpha_k}{\alpha_j} + \alpha_j)}{1 - \frac{\alpha_k}{\alpha_j}} + \langle x_k \frac{\sqrt{\frac{\alpha_k}{\alpha_j}}}{1 - \frac{\alpha_k}{\alpha_j}} + x_0 \frac{\sqrt{\alpha_j}}{1 - \alpha_j}, x_j \rangle\right) \\
&\propto \exp\left(-\frac{1}{2}x_j^T A^{-1}x_j + b^T x_j\right)
\end{aligned}$$

where $A^{-1} = \frac{(\frac{\alpha_k}{\alpha_j} + \alpha_j)}{1 - \frac{\alpha_k}{\alpha_j}}$ and $b = x_k \frac{\sqrt{\frac{\alpha_k}{\alpha_j}}}{1 - \frac{\alpha_k}{\alpha_j}} + x_0 \frac{\sqrt{\alpha_j}}{1 - \alpha_j}$. This means that $q_{j|0,k}(\cdot|x_0, x_k) \sim \mathcal{N}(Ab, A)$ and straightforward calculation shows that $A = \sigma_{j|k}^2 I_d$ and $Ab = \mu_{j|0,k}(x_0, x_k)$. $\square$

Now, DDPM uses this distribution to defines the generator as follow :

- Define the **backward transition** by considering the bridge kernel between k and $k + 1$ and replacing x_0 by its approximation knowing x_{k+1} .

$$p_{k|k+1}^\theta(x_k|x_{k+1}) := q_{k|0,k+1}(x_k|\hat{x}_{0|k+1}^\theta(x_{k+1}), x_{k+1})$$

- Take as an **initialisation** $p_{0|1}^\theta(\cdot|x_1) = \delta_{\hat{x}_{0|1}^\theta(x_1)}$
- Define the **generative model** with its distribution $p_{0:n}^\theta$ (this distribution is supposed to be an approximation of p_{data}) as follow:

$$\forall x_{0:n} \in (\mathbb{R}^d)^n, \quad p_{0:n}^{\theta*}(x_{0:n}) = p_n(x_n) \prod_{k=0}^{n-1} p_{k|k+1}^{\theta*}(x_k|x_{k+1})$$

In the following, we will omit the dependence in θ^* for the sake of clarity and note p_k the k -th marginal distribution of $p_{0:n}$. Also, for all $0 \leq l < m \leq n$, we denote by $p_{l|m}(\cdot|x_m)$ the conditional distribution of X_l given $X_m = x_m$ when $(X_0, \dots, X_n) \sim p_{0:n}$.

Assumption 2.5. For all $0 \leq l < m \leq n$ and $x_m \in \mathbb{R}^d$, we have

$$p_{l|m}(x_l|x_m) = \int \prod_{k=l}^{m-1} p_{k|k+1}(x_k|x_{k+1}) dx_{l+1:m-1}.$$

Proof. By definition of p_m , we have

$$\begin{aligned}
p_m(x_m) &= \int p_n(x_n) \prod_{k=0}^{n-1} p_{k|k+1}(x_k|x_{k+1}) dx_{0:m-1} dx_{m+1:n} \\
&= \int p_n(x_n) \prod_{k=m}^n p_{k|k+1}(x_k|x_{k+1}) dx_{m+1:n} \int \prod_{k=0}^{m-1} p_{k|k+1}(x_k|x_{k+1}) dx_{0:m-1} \\
&= \int p_n(x_n) \prod_{k=m}^n p_{k|k+1}(x_k|x_{k+1}) dx_{m+1:n}
\end{aligned}$$

Where we used that $\int \prod_{k=0}^{m-1} p_{k|k+1}(x_k|x_{k+1})dx_{0:m-1} = 1$. By marginalizing over $x_{0:l-1}$, $x_{m+1:n}$ and $x_{l+1:m-1}$, we have

$$\begin{aligned} p_{l,m}(x_l, x_m) &= \int p_n(x_n) \prod_{k=0}^{n-1} p_{k|k+1}(x_k|x_{k+1})dx_{0:l-1}dx_{l+1:m-1}dx_{m+1:n} \\ &= \int p_n(x_n) \prod_{k=m}^n p_{k|k+1}(x_k|x_{k+1})dx_{m+1:n} \int \prod_{k=l}^{m-1} p_{k|k+1}(x_k|x_{k+1})dx_{l+1:m-1} \\ &= p_m(x_m) \int \prod_{k=l}^{m-1} p_{k|k+1}(x_k|x_{k+1})dx_{l+1:m-1}. \end{aligned}$$

Finally, by definition of conditional distributions, we have

$$p_{l|m}(x_l|x_m) = \frac{p_{l,m}(x_l, x_m)}{p_m(x_m)} = \int \prod_{k=l}^{m-1} p_{k|k+1}(x_k|x_{k+1})dx_{l+1:m-1}.$$

□

3 Posterior sampling

3.1 Computing of the posterior distribution using the prior given by the generative model

Thanks to our generative model, we get a **prior** p_0 which is the marginal distribution of X_0 when $(X_0, \dots, X_n) \sim p_{0:n}$. Now, given some observations related to X_0 through a likelihood function g_0 , we want to sample from the posterior distribution $\pi_0 \propto g_0(x_0)p_0(x_0)$. The following assumption shows that the distribution π_0 is the time-zero marginal of a Markov model going backward in time from n to 0.

Assumption 3.1 (Posterior as a Markov backward process). *Let $(g_k^*)_{k=0}^n$ be potentials defined for all $x_k \in \mathbb{R}^d$ by*

$$g_k^*(x_k) = \int g_0(x_0)p_{0|k}(x_0|x_k)dx_0 \quad (5)$$

Then the functions $\pi_{k|k+1}$ defined by

$$\pi_{k|k+1}(x_k|x_{k+1}) = \frac{g_k^*(x_k)}{g_{k+1}^*(x_{k+1})}p_{k|k+1}(x_k|x_{k+1}).$$

are Markov transitions and for all $0 \leq k < m \leq n$ and $x_m \in \mathbb{R}^d$,

$$\pi_0(x_0) = \int \pi_n(x_n) \prod_{k=0}^{n-1} \pi_{k|k+1}(x_k|x_{k+1})dx_{1:n}$$

where $\pi_n(x_n) = \frac{1}{Z}g_n^(x_n)p_n(x_n)$ and Z is a constant such that $\pi_0(x_0) = \frac{1}{Z}g_0(x_0)p_0(x_0)$.*

Proof. First, we show that $\pi_{k|k+1}$ are Markov transitions.

$$\begin{aligned}
g_{k+1}^*(x_{k+1}) &= \int g_0(x_0) p_{0|k+1}(x_0|x_{k+1}) dx_0 \\
&= \int g_0(x_0) \left(\int \prod_{i=0}^k p_{i|i+1}(x_i|x_{i+1}) dx_{1:k} \right) dx_0 \\
&= \int g_0(x_0) \left(\int \left(\int \prod_{i=0}^k p_{i|i+1}(x_i|x_{i+1}) dx_{1:k-1} \right) p_{k|k+1}(x_k|x_{k+1}) dx_k \right) dx_0 \\
&= \int g_0(x_0) \int p_{0|k}(x_0|x_k) p_{k|k+1}(x_k|x_{k+1}) dx_k dx_0 \\
&= \int \left(\int g_0(x_0) p_{0|k}(x_0|x_k) dx_0 \right) p_{k|k+1}(x_k|x_{k+1}) dx_k \\
&= \int g_k^*(x_k) p_{k|k+1}(x_k|x_{k+1}) dx_k
\end{aligned}$$

Hence, for all $x_{k+1} \in \mathbb{R}^d$,

$$\int \pi_{k|k+1}(x_k|x_{k+1}) dx_k = \frac{1}{g_{k+1}^*(x_{k+1})} \int g_k^*(x_k) p_{k|k+1}(x_k|x_{k+1}) dx_k = 1.$$

Now we can make those transitions appear into the distribution π_0 . By definition of p_0 , knowing that $g_0 = g_0^*$ we have

$$\begin{aligned}
\pi_0(x_0) &= \frac{1}{Z} g_0^*(x_0) \int p_n(x_n) \prod_{k=0}^{n-1} p_{k|k+1}(x_k|x_{k+1}) dx_{1:n} \\
&= \frac{1}{Z} \int p_n(x_n) g_n^*(x_n) \prod_{k=0}^{n-1} \left(\frac{g_k^*(x_k)}{g_k^*(x_k)} p_{k|k+1}(x_k|x_{k+1}) \right) dx_{1:n} \\
&= \frac{1}{Z} \int p_n(x_n) g_n^*(x_n) \prod_{k=0}^{n-1} \pi_{k|k+1}(x_k|x_{k+1}) dx_{1:n}
\end{aligned}$$

□

3.2 Existing algorithm

Assumption 3.1 states that in order to sample from the posterior, we need follow the backward Markov process defined by the transitions $\pi_{k|k+1}$. However, those transitions are not tractable because they depend on g_k^* which involves computing an expectation under the posterior distribution $p_{0|k}$ which is high-cost to sample from. The paper presents briefly several works that tried to develop tractable approximations of $p_{0|k}$, DPS (Diffusion Posterior Sampling) and Π GDM (Pseudoinverse Gaussian Diffusion Model). They make the following approximations:

$$\begin{aligned}
(DPS) \quad & p_{\hat{0}|k}^{DPS}(\cdot|x_k) := \delta_{x_{\hat{0}|k}(x_k)} \\
(\Pi GDM) \quad & p_{\hat{0}|k}^{\Pi GDM}(\cdot|x_k) := \mathcal{N}(x_{\hat{0}|k}(x_k), (1 - \alpha_k)I_d).
\end{aligned}$$

Then at each step, given a sample X_{k+1} , they want to sample X_k from $\pi_{k|k+1}$. To do so, they observe that the score function of $\pi_{k|k+1}$ can be written as follows:

$$\nabla_{x_k} \log \pi_{k|k+1}(x_k|x_{k+1}) = \nabla \log g_k^*(x_k) + \nabla_{x_k} \log p_{k|k+1}(x_k|x_{k+1}) \quad (6)$$

Hence, they sample from $p_{k|k+1}(\cdot|x_{k+1})$ and then make a correction using one langevin step with the score function of $\hat{g}_k^*$. Where

$$\hat{g}_k^*(x_k) = \int g_0(x_0) p_{0|k}(x_0|x_k) dx_0.$$

is the approximation of the score function of g_k^* using the approximation $p_{0|k}$ of $p_{0|k}$.

Assumption 3.2. Let $\hat{g}_k^*(\cdot)$ be the approximation of $g_k^*(\cdot)$ defined by

$$\hat{g}_k^*(x_k) = \int g_0(x_0) p_{0|k}(x_0|x_k) dx_0.$$

Then for DPS and Π GDM, we have respectively:

$$\begin{aligned} (DPS) \quad & \nabla_{x_k} \log \hat{g}_k^*(x_k) = \nabla_{x_k} \log g_0(\hat{x}_{0|k}(x_k)) \\ (\Pi GDM) \quad & \nabla_{x_k} \log \hat{g}_k^*(x_k) = \mathbb{E}_{Z \sim \mathcal{N}(\hat{x}_{0|k}(x_k), (1-\alpha_k)I_d)} [\nabla_{x_k} \log g_0(Z)] \cdot \nabla_{x_k} \hat{x}_{0|k}(x_k) \end{aligned}$$

Now we can describe the algorithm with the following pseudo-code:

4 The DCPS algorithm

4.1 Intermediate approximations are better than the initial approximation

The main problem of DPS is that the approximation of $p_{0|k}$ is bad at the beginning, when k is big. to tackle this problem, the paper observe that it's possible to make great approximation of the intermediate distributions $p_{l|m}$ for l and m close enough. They "divide" the problem of approximating $p_{0|k}$ into several easier sub-problems. The main result is as follows:

Theorem 4.1. Let $0 \leq l < m \leq n$ and $x_m \in \mathbb{R}^d$. Let $p_{0|k}(\cdot)$ be an any approximation of $p_{0|k}(\cdot|x_k)$. Let $p_{l|k}(\cdot|x_k)$ be an approximation of $p_{l|k}(\cdot|x_k)$ defined by

$$p_{l|k}(x_l|x_k) = \int q_{l|0,k}(x_l|x_0, x_k) p_{0|k}(x_0) dx_0.$$

Assume that $q_{k|k+1}(x_k, x_{k+1}) = p_{k|k+1}(x_k|x_{k+1})$ for all $0 \leq k < n$. Then, for all $l \in \{0, \dots, k-1\}$ and $x_k \in \mathbb{R}^d$,

$$W_2(p_{l|k}(\cdot|x_k), p_{l|k}(\cdot|x_k)) \leq \frac{\sqrt{\alpha_l}(1 - \alpha_k/\alpha_l)}{1 - \alpha_k} W_2(p_{0|k}(\cdot), p_{0|k}(\cdot))$$

4.2 Application to DPS approximation

Application 4.1 (Intermediate distribution using DPS's approximation). *Let $p_{0|k}(\cdot|x_k) = \delta_{x_{0|k}(x_k)}$ be the DPS approximation of $p_{0|k}(\cdot|x_k)$. Let $p_{l|k}(\cdot|x_k)$ be the approximation of $p_{l|k}(\cdot|x_k)$ defined in the preceeding theorem. Then, for all $0 \leq l < k \leq n$ and $x_k \in \mathbb{R}^d$,*

$$p_{l|k}(\cdot|x_k) = \mathcal{N}(\mu_{l|0,k}(x_{0|k}(x_k), x_k), \sigma_{l|k}^2 I_d).$$

Proof.

$$\begin{aligned} p_{l|k}(x_l|x_k) &= \int q_{l|0,k}(x_l|x_0, x_k) p_{0|k}(x_0|x_k) dx_0 \\ &= \int q_{l|0,k}(x_l|x_0, x_k) \delta_{x_{0|k}(x_k)}(x_0) dx_0 \\ &= q_{l|0,k}(x_l|x_{0|k}(x_k), x_k) \end{aligned}$$

And Assumption 2.4 concludes the proof. $\square$

Application 4.2 (Wasserstein distance using DPS approximation). *Let the hypotheses of the previous theorem be satisfied. Let $X_{0|k}$ be any random variable distributed according to $p_{0|k}(\cdot|x_k)$. Let $p_{0|k}(\cdot|x_k) = \delta_{x_{0|k}(x_k)}$ be the DPS approximation of $p_{0|k}(\cdot|x_k)$. Then, for all k and $x_k \in \mathbb{R}^d$,*

$$\begin{aligned} W_2(p_{0|k}(\cdot|x_k), p_{0|k}(\cdot|x_k)) &= \mathbb{E}(\|x_{0|k}(x_k) - X_{0|k}\|^2) \\ W_2(p_{l|k}(\cdot|x_k), p_{l|k}(\cdot|x_k)) &\leq \lambda_{l,k} \mathbb{E}(\|x_{0|k}(x_k) - X_{0|k}\|) \end{aligned}$$

Proof of Application 5.2. Let $X_{0|k} \sim p_{0|k}(\cdot|x_k)$ and $\hat{X}_{0|k} \sim \hat{p}_{0|k}(\cdot|x_k)$. By the definition of the Wasserstein-2 distance:

$$W_2^2(\hat{p}_{0|k}, p_{0|k}) = \inf_{\gamma \in \Gamma(\hat{p}_{0|k}, p_{0|k})} \mathbb{E}_{(X,Y) \sim \gamma} [\|X - Y\|^2]$$

Since $\hat{p}_{0|k} = \delta_{\hat{x}_{0|k}(x_k)}$, the only possible coupling is the product measure, and the distance simplifies to:

$$W_2^2(\hat{p}_{0|k}, p_{0|k}) = \mathbb{E}_{X_{0|k} \sim p_{0|k}} [\|\hat{x}_{0|k}(x_k) - X_{0|k}\|^2].$$

Now, consider the intermediate variables $X_{l|k}$ and $\hat{X}_{l|k}$. They can be constructed as:

$$\begin{aligned} X_{l|k} &= \mu_{l|0,k}(X_{0|k}, x_k) + \sigma_{l|k} \epsilon \\ \hat{X}_{l|k} &= \mu_{l|0,k}(\hat{x}_{0|k}(x_k), x_k) + \sigma_{l|k} \epsilon \end{aligned}$$

where $\epsilon \sim \mathcal{N}(0, I_d)$. This construction defines a coupling between $\hat{p}_{l|k}$ and $p_{l|k}$. Using the linearity of $\mu_{l|0,k}$ with respect to x_0 :

$$\begin{aligned} \hat{X}_{l|k} - X_{l|k} &= \mu_{l|0,k}(\hat{x}_{0|k}(x_k), x_k) - \mu_{l|0,k}(X_{0|k}, x_k) \\ &= \lambda_{l,k}(\hat{x}_{0|k}(x_k) - X_{0|k}). \end{aligned}$$

Thus, for this specific coupling, we have:

$$\mathbb{E}[\|\hat{X}_{l|k} - X_{l|k}\|^2] = \lambda_{l,k}^2 \mathbb{E}[\|\hat{x}_{0|k}(x_k) - X_{0|k}\|^2].$$

Since W_2^2 is the infimum over all couplings:

$$W_2^2(\hat{p}_{l|k}, p_{l|k}) \leq \mathbb{E}[\|\hat{X}_{l|k} - X_{l|k}\|^2] = \lambda_{l,k}^2 W_2^2(\hat{p}_{0|k}, p_{0|k}).$$

Taking the square root and using $\lambda_{l,k} = \frac{\sqrt{\alpha_l(1-\alpha_k/\alpha_l)}}{1-\alpha_k}$, we obtain the result. $\square$

4.3 Application to Π GDM approximation

Application 4.3 (Intermediate distribution using Π GDM's approximation). *Let $p_{\hat{0}|k}(\cdot|x_k) = \mathcal{N}(x_{\hat{0}|k}(x_k), (1-\alpha_k)I_d)$ be the Π GDM approximation of $p_{0|k}(\cdot|x_k)$. Let $p_{\hat{l}|k}(\cdot|x_k)$ be the approximation of $p_{l|k}(\cdot|x_k)$ defined in the preceding theorem. Then, for all $0 \leq l < k \leq n$ and $x_k \in \mathbb{R}^d$,*

$$p_{\hat{l}|k}(\cdot|x_k) = \mathcal{N}(\mu_{l|0,k}(x_{\hat{0}|k}(x_k), x_k), (\sigma_{l|k}^2 + \lambda_{l,k}^2(1-\alpha_k))I_d).$$

Proof. for all l, k such that $0 \leq l < k \leq n$ and $x_k \in \mathbb{R}^d$, $\mu_{l|0,k}(x_0, x_k) = \lambda_{l,k}x_0 + \beta_{l,k}x_k$ where $\lambda_{l,k} = \frac{\sqrt{\alpha_l(1-\alpha_k/\alpha_l)}}{1-\alpha_k}$ and $\beta_{l,k} = \frac{\sqrt{\alpha_k/\alpha_l(1-\alpha_l)}}{1-\alpha_k}$. Thus, we can rewrite $q_{l|0,k}(x_l|x_0, x_k)$ as a function of x_0 . Let $x_l, x_k \in \mathbb{R}^d$. For all $x_0 \in \mathbb{R}^d$,

$$\begin{aligned} q_{l|0,k}(x_l|x_0, x_k) &= \frac{1}{(2\pi\sigma_{l|k}^2)^{d/2}} \exp\left(-\frac{1}{2\sigma_{l|k}^2}\|x_l - \mu_{l|0,k}(x_0, x_k)\|^2\right) \\ &= \frac{1}{(2\pi\sigma_{l|k}^2)^{d/2}} \exp\left(-\frac{1}{2\sigma_{l|k}^2}\|x_l - (\lambda_{l,k}x_0 + \beta_{l,k}x_k)\|^2\right) \\ &= \frac{1}{(2\pi\sigma_{l|k}^2)^{d/2}} \exp\left(-\frac{\lambda_{l,k}^2}{2\sigma_{l|k}^2}\left\|x_0 - \frac{x_l - \beta_{l,k}x_k}{\lambda_{l,k}}\right\|^2\right) \\ &= \lambda_{l,k}^{-d} \mathcal{N}\left(x_0; \frac{x_l - \beta_{l,k}x_k}{\lambda_{l,k}}, \frac{\sigma_{l|k}^2}{\lambda_{l,k}^2}I_d\right). \end{aligned}$$

Thus, by reinjecting this expression into the definition of $p_{\hat{l}|k}$, we have

$$\begin{aligned} p_{\hat{l}|k}(x_l|x_k) &= \int q_{l|0,k}(x_l|x_0, x_k) p_{\hat{0}|k}(x_0|x_k) dx_0 \\ &= \lambda_{l,k}^{-d} \int \mathcal{N}\left(x_0; \frac{x_l - \beta_{l,k}x_k}{\lambda_{l,k}}, \frac{\sigma_{l|k}^2}{\lambda_{l,k}^2}I_d\right) \mathcal{N}(x_0; x_{\hat{0}|k}(x_k), (1-\alpha_k)I_d) dx_0. \end{aligned}$$

By using lemma 6.2, we have

$$\begin{aligned} p_{\hat{l}|k}(x_l|x_k) &= \lambda_{l,k}^{-d} \mathcal{N}\left(\frac{x_l - \beta_{l,k}x_k}{\lambda_{l,k}}; x_{\hat{0}|k}(x_k), \left(\frac{\sigma_{l|k}^2}{\lambda_{l,k}^2} + 1 - \alpha_k\right)I_d\right) \\ &= \lambda_{l,k}^{-d} \frac{1}{(2\pi(\frac{\sigma_{l|k}^2}{\lambda_{l,k}^2} + 1 - \alpha_k))^{d/2}} \exp\left(-\frac{\left\|\frac{x_l - \beta_{l,k}x_k}{\lambda_{l,k}} - x_{\hat{0}|k}(x_k)\right\|^2}{2(\frac{\sigma_{l|k}^2}{\lambda_{l,k}^2} + 1 - \alpha_k)}\right) \\ &= \frac{1}{(2\pi(\sigma_{l|k}^2 + \lambda_{l,k}^2(1-\alpha_k)))^{d/2}} \exp\left(-\frac{\|x_l - (\lambda_{l,k}x_{\hat{0}|k}(x_k) + \beta_{l,k}x_k)\|^2}{2(\sigma_{l|k}^2 + \lambda_{l,k}^2(1-\alpha_k))}\right). \end{aligned}$$

Furthermore,

$$\begin{aligned}
\sigma_{l|k}^2 + \lambda_{l,k}^2(1 - \alpha_k) &= \frac{(1 - \alpha_l)(1 - \alpha_k/\alpha_l)}{1 - \alpha_k} + \frac{\alpha_l(1 - \alpha_k/\alpha_l)^2}{(1 - \alpha_k)^2}(1 - \alpha_k) \\
&= \frac{(1 - \alpha_l)(1 - \alpha_k/\alpha_l) + \alpha_l(1 - \alpha_k/\alpha_l)^2}{1 - \alpha_k} \\
&= \frac{1 - \alpha_k/\alpha_l}{1 - \alpha_k} ((1 - \alpha_l) + \alpha_l(1 - \alpha_k/\alpha_l)) \\
&= (1 - \alpha_k/\alpha_l).
\end{aligned}$$

and by definition of $\mu_{l|0,k}$, we have

$$\lambda_{l,k}x_{\hat{0}|k}(x_k) + \beta_{l,k}x_k = \mu_{l|0,k}(x_{\hat{0}|k}(x_k), x_k)$$

which leads to the density of the expected gaussian distribution. $\square$

The following application consists in proving the theorem in the particular case where the approximation of $p_{0|k}$ is given by the ΠGDM approximation.

Application 4.4 (Wasserstein distance using ΠGDM 's approximation). *Let the hypotheses of the previous theorem be satisfied. Let $X_{0|k}$ be any random variable distributed according to $p_{0|k}(\cdot|x_k)$. Let $p_{\hat{0}|k}(\cdot|x_k) = \mathcal{N}(x_{\hat{0}|k}(x_k), (1 - \alpha_k)I_d)$ be the ΠGDM approximation of $p_{0|k}(\cdot|x_k)$. Then, for all k and $x_k \in \mathbb{R}^d$,*

$$\begin{aligned}
W_2(p_{\hat{0}|k}(\cdot|x_k), p_{0|k}(\cdot|x_k)) &= \mathbb{E}(\|x_{\hat{0}|k}(x_k) - X_{0|k}\|^2) + d(1 - \alpha_k) \\
W_2(p_{\hat{l}|k}(\cdot|x_k), p_{l|k}(\cdot|x_k)) &\leq \lambda_{l,k}\mathbb{E}(\|x_{\hat{0}|k}(x_k) - X_{0|k}\|) + d(\sqrt{\sigma_{l|k}^2 + \lambda_{l,k}^2(1 - \alpha_k)} - \sigma_{l|k})^2
\end{aligned}$$

Proof. Let's define two random variables $X_{l|k}$ and $\hat{X}_{l|k}$ such that

$$\begin{aligned}
X_{l|k} &= \lambda_{l,k}X_{0|k} + \beta_{l,k}x_k + \sigma_{l|k}\epsilon, \\
\hat{X}_{l|k} &= \lambda_{l,k}x_{\hat{0}|k}(x_k) + \beta_{l,k}x_k + \sqrt{\sigma_{l|k}^2 + \lambda_{l,k}^2(1 - \alpha_k)}\epsilon.
\end{aligned}$$

where $\epsilon \sim \mathcal{N}(0, I_d)$. The beginning of the proof of proposition A1 in the paper shows that $X_{l|k} \sim p_{l|k}(\cdot|x_k)$. Our preceding application shows that $\hat{X}_{l|k} \sim p_{\hat{l}|k}(\cdot|x_k)$.

Using the preceding coupling and this lemma conditionnaly to $X_{0|k}$, we have

$$\begin{aligned}
&W_2^2(p_{\hat{l}|k}(\cdot|x_k), p_{l|k}(\cdot|x_k)) \\
&= \inf_{U \sim p_{\hat{l}|k}(\cdot|x_k), V \sim p_{l|k}(\cdot|x_k)} \mathbb{E}[\|U - V\|^2] \\
&\leq \mathbb{E}[\|\hat{X}_{l|k} - X_{l|k}\|^2] \\
&= \mathbb{E}[\|\lambda_{l,k}x_{\hat{0}|k}(x_k) - \lambda_{l,k}X_{0|k}\|^2] + d\left(\sqrt{\sigma_{l|k}^2 + \lambda_{l,k}^2(1 - \alpha_k)} - \sigma_{l|k}\right)^2 \\
&= \lambda_{l,k}^2\mathbb{E}[\|x_{\hat{0}|k}(x_k) - X_{0|k}\|^2] + d\left(\sqrt{\sigma_{l|k}^2 + \lambda_{l,k}^2(1 - \alpha_k)} - \sigma_{l|k}\right)^2.
\end{aligned}$$

Furthermore, $\hat{X}_{0|k} \sim p_{\hat{0}|k}(\cdot|x_k) = \mathcal{N}(x_{\hat{0}|k}(x_k), (1 - \alpha_k)I_d)$, hence using again the previous lemma, with gaussian $\hat{X}_{0|k}$ and constant $X_{0|k}$ conditionally to $X_{0|k}$, we have

$$\begin{aligned}
\mathbb{E}[\|\hat{X}_{0|k} - X_{0|k}\|^2] &= \mathbb{E}\left[\mathbb{E}[\|\hat{X}_{0|k} - X_{0|k}\|^2 | X_{0|k}]\right] \\
&= \mathbb{E}[\|x_{\hat{0}|k}(x_k) - X_{0|k}\|^2] + d(1 - \alpha_k).
\end{aligned}$$

□

Those exact computations confirm the proposition of the paper. Indeed,

$$\left(\sqrt{\sigma_{l|k}^2 + \lambda_{l,k}^2(1 - \alpha_k)} - \sigma_{l|k}\right)^2 \leq \lambda_{l,k}^2(1 - \alpha_k)$$

which gives

$$\begin{aligned} W_2^2(p_{l|k}(\cdot|x_k), p_{l|k}(\cdot|x_k)) &\leq \lambda_{l,k}^2 \mathbb{E} [\|x_{0|k}(x_k) - X_{0|k}\|^2] + d\lambda_{l,k}^2(1 - \alpha_k) \\ &= \lambda_{l,k}^2 W_2^2(p_{0|k}(\cdot|x_k), p_{0|k}(\cdot|x_k)) \end{aligned}$$

4.4 Intermediate posterior

Now as before let g_0 be a likelihood function. Let $L < n$ and $(k_l)_{l=0}^L$ be an increasing sequence of integers such that $k_0 = 0$ and $k_L = n$. The goal of the DCPS algorithm is to divide the solution provided by DPS into several sub-problems in order to benefit from the preceding result.

- DPS and IIGDM try to sample π_0 directly using n backward transitions $\pi_{m|m+1}$ that use $p_{0|m}$, g_0 , and $p_{m|m+1}$.
- DCPS decompose the problem into L sub-problems. It separates $0, \dots, n$ into L blocks $\{k_l + 1, \dots, k_{l+1}\}$. For each block $l \in \{0, \dots, L\}$ we try to sample $\pi(k_l)$ using $k_{l+1} - k_l$ backward transitions $\pi_{m|m+1}^l$ that use $p_{k_l|m}$, g_0 , and $p_{m|m+1}$.

DCPS uses a different likelihood function for each block. For all $0 \leq l < L$, the paper defines g_{k_l} the likelihood by

$$g_{k_l}(x) = g_0(x/\beta_{k_l})_{k_l}^\gamma$$

where $\gamma_{k_l}, \beta_{k_l} > 0$ are two constants. Thanks to those likelihoods, we can express the posterior of each block as a Markov backward process.

Assumption 4.1 (Block Posterior as a Markov backward process). *Let $(g_m^{l,*})_{m=k_l+1}^{k_{l+1}}$ be potentials defined for all $x_m \in \mathbb{R}^d$ by*

$$g_m^{l,*}(x_m) = \int g_{k_l}(x_m) p_{k_l|m}(x_{k_l}|x_m) dx_{k_l} \quad (7)$$

Then for all $m \in \{k_l + 1, \dots, k_{l+1}\}$, the functions $\pi_{m|m+1}^l$ defined by

$$\pi_{m|m+1}^l(x_m|x_{m+1}) = \frac{g_m^{l,*}(x_m)}{g_{m+1}^{l,*}(x_{m+1})} p_{m|m+1}(x_m|x_{m+1}).$$

are Markov transitions and for all $0 \leq k < m \leq n$ and $x_m \in \mathbb{R}^d$,

$$\pi_{k_l}(x_{k_l}) = \int \pi_{k_{l+1}}^l(x_{k_{l+1}}) \prod_{k=k_l}^{k_{l+1}-1} \pi_{k|k+1}^l(x_k|x_{k+1}) dx_{k_{l+1}}$$

where $\pi_{k_{l+1}}^l(x_{k_{l+1}}) = \frac{1}{Z_{k_l}} g_{k_{l+1}}^{l,}(x_{k_{l+1}}) p_{k_{l+1}}(x_{k_{l+1}})$ with Z_{k_l} a normalizing constant.*

Proof. Similar to proof of Assumption 3.1 □

Now for each block the algorithm of dcps proceed as follow :

- Get a sample from $X_{k_{l+1}} \sim \pi_{k_{l+1}}$.
- **Step 1** Perform several Langevin steps starting from this sample $X_{k_{l+1}}$ in order to match the distribution $\pi_{k_{l+1}}^l$. At the end of this step, we get $X_{k_{l+1}}^l \sim \pi_{k_{l+1}}^l$.
- **Step 2** For each m starting from $k_{l+1} - 1$ to k_l , get a sample $X_m^l \sim \pi_{m|m+1}^l(\cdot | X_{m+1}^l)$
- Define $X_{k_l} = X_{k_l}^l$
- Do it again until $l = 0$

4.5 Step 1 : Langevin

Algorithm 1 DCPS Step 1: Langevin guidance for block l

Require: Sample $X_{k_{l+1}} \sim \pi_{k_{l+1}}$, block index l , steps M , step size γ .

```

1:  $\tilde{X}_0 \leftarrow X_{k_{l+1}}$ 
2: for  $j = 0$  to  $M - 1$  do
3:   Draw  $Z_j \sim \mathcal{N}(0, I_d)$ 
4:   Draw  $Z'_j \sim \mathcal{N}(0, I_d)$ 
5:    $s_2 \leftarrow \hat{s}_{k_{l+1}}(\tilde{X}_j)$ 
6:    $s_1 \leftarrow \nabla \log g_{k_l}(\mu_{k_l|k_{l+1}}(\tilde{X}_j) + \sigma_{k_l|k_{l+1}} Z'_j)$ 
7:    $G_\gamma^l \leftarrow \frac{s_1 + s_2}{1 + \gamma \|s_1 + s_2\|}$ 
8:    $\tilde{X}_{j+1} \leftarrow \tilde{X}_j + \gamma G_\gamma^l + \sqrt{2\gamma} Z_{j+1}$ 
9: end for
10:  $X_{k_{l+1}}^l \leftarrow \tilde{X}_M$ 
11: Return  $X_{k_{l+1}}^l$ 

```

We want to perform step 1 of block $l \in \{1, \dots, L\}$. Thus, we have at our disposal a sample $X_{k_{l+1}} \sim \pi_{k_{l+1}}$. We want to modify this sample for it to match the distribution $\pi_{k_{l+1}}^l$ (wich is specific to block l). For that the paper does M steps of TULA (Tamed Unadjusted Langevin Algorithm). They define $(\tilde{X}_j)_{m=0}^M$ as follows:

$$\tilde{X}_0 = X_{k_{l+1}}, \quad \tilde{X}_{j+1} = \tilde{X}_j + \delta G_\gamma^l(\tilde{X}_j) + \sqrt{2\delta} Z_{j+1} \quad (8)$$

where $G_\gamma^l(x) = \nabla \log \hat{\pi}_{k_{l+1}}^l(\tilde{X}_j) / (1 + \delta \|\nabla \log \hat{\pi}_{k_{l+1}}^l(\tilde{X}_j)\|)$ and $Z_{j+1} \sim \mathcal{N}(0, I_d)$. Because of the expression of $\pi_{k_{l+1}}^l$, we have

$$\nabla \log \pi_{k_{l+1}}^l(x) = \nabla \log g_{k_{l+1}}^{l,*}(x) + \nabla \log p_{k_{l+1}}(x) \quad (9)$$

To run this step, we need two approximations :

$$\begin{aligned} \nabla \log g_{k_{l+1}}^{l,*}(x) &\approx \nabla \log \hat{g}_{k_{l+1}}^{l,*}(x) \\ \nabla \log p_{k_{l+1}}(x) &\approx \hat{s}_{k_{l+1}}(x) \end{aligned}$$

where $\hat{g}_{k_{l+1}}^{l,\star}(x) = \int g_{k_l}(x_{k_l})p_{k_l|k_{l+1}}(\hat{x}_{k_l}|x)dx_{k_l}$ is an approximation of $g_{k_{l+1}}^{l,\star}(x)$ using an approximation $p_{k_l|\hat{k}_{l+1}}$ of $p_{k_l|k_{l+1}}$. And $\hat{s}_{k_{l+1}}(x)$ is the score model resulting from pre-training. But instead of computing $\nabla \log \hat{g}_{k_{l+1}}^{l,\star}(x) + \hat{s}_{k_{l+1}}(x)$ directly, which is often hard because it needs to compute an expectation, they make what they call a classic approximation of the expectation by a sample. Indeed, they observe that

$$\hat{g}_m^{l,\star}(x_m) = \mathbb{E}_{U \sim \hat{p}_{k_l|m}(\cdot|x_m)} [g_{k_l}(U)] \approx g_{k_l}(U) \quad U \sim \hat{p}_{k_l|m}(\cdot|x_m) \quad (10)$$

Thus, they approximate $\nabla \log \hat{g}_{k_{l+1}}^{l,\star}(x)$ by

$$\nabla \log g_{k_l}(U) \quad U \sim \hat{p}_{k_l|k_{l+1}}(\cdot|x) \quad (11)$$

But $\hat{p}_{k_l|k_{l+1}}(\cdot|x) = \mathcal{N}(\mu_{k_l|k_{l+1}}(x), \sigma_{k_l|k_{l+1}}^2 I_d)$ reinjecting in the expression of $G_\gamma^l(x)$ we get an approximation $\tilde{G}_\gamma^l(x)$ of $G_\gamma^l(x)$ defined by

$$\tilde{G}_\gamma^l(x) = \frac{\nabla \log g_{k_l}(\mu_{k_l|k_{l+1}}(x) + \sigma_{k_l|k_{l+1}} Z) + \hat{s}_{k_{l+1}}(x)}{1 + \gamma \|\nabla \log g_{k_l}(\mu_{k_l|k_{l+1}}(x) + \sigma_{k_l|k_{l+1}} Z) + \hat{s}_{k_{l+1}}(x)\|} \quad Z \sim \mathcal{N}(0, I_d) \quad (12)$$

This expression leads to the algorithm 1.

4.6 Step 2 : Backward Sampling

Algorithm 2 DCPS Step 2: Backward Sampling on Block l

Require: Sample $X_{k_{l+1}}^l \sim \pi_{k_{l+1}}^l$, block index l , step size ζ .

```

1: for  $m = k_{l+1} - 1$  downto  $k_l$  do
2:   Initialize  $(\mu_{m,0}^l, \nu_{m,0}^l)$ 
3:   for  $i = 0$  to  $K - 1$  do
4:     Draw  $Z_i \sim \mathcal{N}(0, I_d)$ 
5:     Draw  $U_i \sim \hat{p}_{k_l|m}(\mu_{m,i}^l + e^{\nu_{m,i}^l/2} \odot Z_i)$ 
6:      $g \leftarrow \nabla_{(\mu,\nu)} L_m^l(\mu_{m,i}^l, \nu_{m,i}^l, X_{m+1}^l, Z_i, U_i)$ 
7:      $(\mu_{m,i+1}^l, \nu_{m,i+1}^l) \leftarrow (\mu_{m,i}^l, \nu_{m,i}^l) - \zeta \frac{g}{\|g\|}$ 
8:   end for
9:    $\hat{\mu}_m^l \leftarrow \mu_{m,K}^l, \hat{\sigma}_m^l = e^{\nu_{m,K}^l/2}$ 
10:  Draw  $X_m^l \sim \mathcal{N}(\hat{\mu}_m^l, \text{diag}((\hat{\sigma}_m^l)^2))$ 
11: end for
12:  $X_{k_l} \leftarrow X_{k_l}^l$ 
13: Return  $X_{k_l}$ 
```

For the second step of block $l \in \{1, \dots, L\}$, we have at our disposal a sample $X_{k_{l+1}}^l$. Starting from $m = k_{l+1} - 1$ to k_l , suppose that we have already a sample X_{m+1}^l . Then we want to sample X_m^l from $\pi_{m|m+1}^l(\cdot|X_{m+1}^l)$. But instead of sampling directly from $\pi_{m|m+1}^l(\cdot|X_{m+1}^l)$, we approximate this distribution by a gaussian distribution $\mathcal{N}(\hat{\mu}, \text{diag}(\hat{\sigma}^2))$ whose parameters minimize the KL divergence between $\pi_{m|m+1}^l(\cdot|X_{m+1}^l)$ and $\mathcal{N}(\hat{\mu}, \text{diag}(\hat{\sigma}^2))$. To solve this optimization problem, we can express it as follows:

Assumption 4.2. For all $m \in \{k_l, \dots, k_{l+1} - 1\}, x \in \mathbb{R}^d$,

$$\min_{(\mu, \sigma) \in \mathbb{R}^d \times \mathbb{R}_+^d} KL(\mathcal{N}(\mu, \text{diag}(e^\sigma)) || \pi_{m|m+1}^l(\cdot | X_{m+1}^l)) = \min_{(\mu, \nu) \in \mathbb{R}^{2d}} \mathcal{L}_m^l(\mu, \nu, x)$$

where for all $(\mu, \nu) \in \mathbb{R}^{2d}$,

$$\begin{aligned} \mathcal{L}_m^l(\mu, \nu, x) = & \frac{\|\mu - q_{m|m+1}(X_{m+1}^l)\|^2}{2\sigma_{m|m+1}^2} - \frac{1}{2} \sum_{i=1}^d \left(\frac{e^{\nu_i}}{\sigma_{m|m+1}^2} - \nu_i \right) \\ & - \mathbb{E}_{Z \sim \mathcal{N}(0, I_d)} [\log(g_m^{l,*}(\mu + e^{\nu/2} \odot Z))] + C \end{aligned}$$

Proof. First, we can observe that $\{(\mu, \sigma) \in \mathbb{R}^d \times \mathbb{R}_+^d\}$ is equal to $\{(\mu, e^\nu) \in \mathbb{R}^{2d}\}$. Thus, we can rewrite the KL divergence as follows.

$$\begin{aligned} & KL(\mathcal{N}(\mu, \text{diag}(e^\sigma)) || \pi_{m|m+1}^l(\cdot | X_{m+1}^l)) \\ &= \int \mathcal{N}(\mu, \text{diag}(e^\sigma)) \log \left(\frac{\mathcal{N}(\mu, \text{diag}(e^\sigma))}{p_{m|m+1}(x_m | X_{m+1}^l)} \right) dx_m - \int \mathcal{N}(\mu, \text{diag}(e^\sigma)) \log \left(\frac{g_m^{l,*}(x_m)}{g_{m+1}^{l,*}(X_{m+1}^l)} \right) dx_m \\ &= KL(\mathcal{N}(\mu, \text{diag}(e^\sigma)) || p_{m|m+1}(\cdot | X_{m+1}^l)) - \mathbb{E}_{Z \sim \mathcal{N}(\mu, \text{diag}(e^\sigma))} [\log(g_m^{l,*}(Z))] + C \end{aligned}$$

We already know that $p_{m|m+1}(\cdot | X_{m+1}^l) = \mathcal{N}(q_{m|m+1}(X_{m+1}^l), \sigma_{m|m+1}^2 I_d)$. Thus, using lemma 6.4, the first term of the sum has an explicit expression :

$$\begin{aligned} & KL(\mathcal{N}(\mu, \text{diag}(e^\sigma)) || p_{m|m+1}(\cdot | X_{m+1}^l)) \\ &= \frac{1}{2} \sum_{i=1}^d \left[\frac{e^{\sigma_i}}{\sigma_{m|m+1}^2} + \frac{(\mu_i - q_{m|m+1}(X_{m+1}^l)_i)^2}{\sigma_{m|m+1}^2} - 1 - \sigma_i + \log(\sigma_{m|m+1}^2) \right] \\ &= \frac{\|\mu - q_{m|m+1}(X_{m+1}^l)\|^2}{2\sigma_{m|m+1}^2} - \frac{1}{2} \sum_{i=1}^d \left(\frac{e^{\sigma_i}}{\sigma_{m|m+1}^2} - \sigma_i \right) + C \end{aligned}$$

Reinjecting this expression into the previous equation concludes the proof. $\square$

Instead of using the expectation $\mathbb{E}_{Z \sim \mathcal{N}(0, I_d)} [\log(g_m^{l,*}(\mu + e^{\nu/2} \odot Z))]$ in the loss function, As before, we observe that

$$\hat{g}_m^{l,*}(x_m) = \mathbb{E}_{U \sim \hat{p}_{k_l|m}(|x_m)} [g_{k_l}(U)] \approx g_{k_l}(U) \quad U \sim \hat{p}_{k_l|m}(\cdot | x_m) \quad (13)$$

where in the last line they approximate the expectation by a single sample. Thus, $\mathbb{E} \log(g_m^{l,*}(\mu + e^{\nu/2} \odot Z)) \approx \log(g_{k_l}(U))$ where $U \sim \hat{p}_{k_l|m}(|\mu + e^{\nu/2} \odot Z)$ and by again taking a single sample as the approximation of the expectation, we have

$$\begin{aligned} & \mathbb{E}_{Z \sim \mathcal{N}(0, I_d)} [\log(g_m^{l,*}(\mu + e^{\nu/2} \odot Z))] \approx \log(g_{k_l}(U)) \\ & \text{where } Z \sim \mathcal{N}(0, I_d) \text{ and } U | Z \sim \hat{p}_{k_l|m}(|\mu + e^{\nu/2} \odot Z) \end{aligned}$$

Thus, for all $(\mu, \nu) \in \mathbb{R}^{2d}, x \in \mathbb{R}^d, Z \sim \mathcal{N}(0, I_d)$, and $U | Z \sim \hat{p}_{k_l|m}(|\mu + e^{\nu/2} \odot Z)$, we define the approximated loss function :

$$L_m^l(\mu, \nu, x, Z, U) = \frac{\|\mu - q_{m|m+1}(x)\|^2}{2\sigma_{m|m+1}^2} - \frac{1}{2} \sum_{i=1}^d \left(\frac{e^{\nu_i}}{\sigma_{m|m+1}^2} - \nu_i \right) - \log(g_{k_l}(U)) \quad (14)$$

Let R_m^l be the distribution of the couple (Z, U) where $Z \sim \mathcal{N}(0, I_d)$ and $U \sim \hat{p}_{k_l|m}(|\mu + e^{\nu/2} \odot Z|)$. The algorithm perform K steps of a stochastic gradient descent on this loss function. Thus, they start from some $(\mu_{m,0}^l, \nu_{m,0}^l)$, and for all $i \in \{0, \dots, K-1\}$, they sample $(Z_i, U_i) \sim R_m^l$ and perform the update :

$$(\mu_{m,i+1}^l, \nu_{m,i+1}^l) = (\mu_{m,i}^l, \nu_{m,i}^l) - \zeta \frac{\nabla_{(\mu,\nu)} L_m^l(\mu_{m,i}^l, \nu_{m,i}^l, X_{m+1}^l, Z_i, U_i)}{\|\nabla_{(\mu,\nu)} L_m^l(\mu_{m,i}^l, \nu_{m,i}^l, X_{m+1}^l, Z_i, U_i)\|} \quad (15)$$

where $\zeta > 0$ is the step size of the stochastic gradient descent. At the end of those K steps, they define

$$\hat{\mu}_m^l = \mu_{m,K}^l, \quad \hat{\sigma}_m^l = e^{\nu_{m,K}^l/2} \quad (16)$$

and sample $X_m^l \sim \mathcal{N}(\hat{\mu}_m^l, \text{diag}((\hat{\sigma}_m^l)^2))$. This leads to the algorithm 2.

5 DCPS for linear inverse problems

In this section, we will prove results mentions in the paper of particular use of DCPS. In all this section, we suppose we use the DPS approximation of $p_{0|k}(\cdot|x_k)$, ie $p_{\hat{0}|k}(x|x_k) = \delta_{x=x_{\hat{0}|k}(x_k)}$ wich gives the approximation

$$p_{k_l|m}(x_{k_l}|x_m) = \mathcal{N}(x_{k_l}; \mu_{k_l|m}(x_m), (1 - \alpha_{k_l})I_d)$$

where we note $\mu_{k_l|m}(x_m) := \mu_{k_l|0,m}(x_{\hat{0}|m}(x_m), x_m)$. We will use this notation in all this section. Furthermore, we only consider the case of a linear inverse problems, wich means we define the likelihood g_0 as follows:

$$g_0(x) = \mathcal{N}(y; Ax, \sigma_y^2 I)$$

. where we fix $d_y \in \mathbb{N}, y \in \mathbb{R}^{d_y}, \sigma_y > 0$

5.1 Potentials in the case of linear inverse problems

Assumption 5.1. Let g_0 be a likelihood function defined for all $x \in \mathbb{R}^d$ by

$$g_0(x) = \mathcal{N}(y; Ax, \sigma_y^2 I)$$

and g_{k_l} be the likelihood defined for all $x \in \mathbb{R}^d$ by

$$g_{k_l}(x) = g_0(x/\beta_{k_l})^{\gamma_{k_l}}$$

where $(\beta_{k_l}, \gamma_{k_l}) = (\sqrt{\alpha_{k_l}}, \alpha_{k_l})$. Then, for all $m \in \{k_l + 1, \dots, k_{l+1}\}$ and $x_m \in \mathbb{R}^d$,

$$g_{k_l}(x) \propto \mathcal{N}(\sqrt{\alpha_{k_l}}y; Ax, \sigma_y^2 I) \quad (17)$$

$$g_m^{l,*}(x_m) = \mathcal{N}(\sqrt{\alpha_{k_l}}y; A\mu_{k_l|m}(x_m), \Sigma_{k_l|m}) \quad (18)$$

where $\mu_{k_l|m}(x_m) = \beta_{k_l}x_{\hat{0}|m}(x_m)$ and $\Sigma_{k_l|m} = \sigma_y^2 I + \frac{\gamma_{k_l}}{d} \|A\|_F^2 (1 - \alpha_{k_l}) I_d$.

Proof. First, we can rewrite g_{k_l} as follows :

$$\begin{aligned}
g_{k_l}(x) &= g_0(x/\beta_{k_l})^{\gamma_{k_l}} \\
&= \left(\frac{1}{(2\pi\sigma_y^2)^{d/2}} \exp\left(-\frac{\|y - A(x/\beta_{k_l})\|^2}{2\sigma_y^2}\right) \right)^{\gamma_{k_l}} \\
&= \frac{1}{(2\pi\sigma_y^2)^{d\gamma_{k_l}/2}} \exp\left(-\frac{\alpha_{k_l}\|y - (A/\sqrt{\alpha_{k_l}})x\|^2}{2\sigma_y^2}\right) \\
&= \frac{1}{(2\pi\sigma_y^2)^{d\gamma_{k_l}/2}} \exp\left(-\frac{\|\sqrt{\alpha_{k_l}}y - Ax\|^2}{2\sigma_y^2}\right)
\end{aligned}$$

which gives the first result. Then for all x ,

$$g_{k_l}(x) \propto \exp\left(-\frac{1}{2\sigma_y^2}(\|\sqrt{\alpha_{k_l}}y - Ax\|^2)\right) \quad (19)$$

$$\propto \exp\left(-\frac{1}{2\sigma_y^2}(x^T A^T A x - 2\sqrt{\alpha_{k_l}}y^T A x)\right) \quad (20)$$

$$\propto \exp\left(-\frac{1}{2}x^T \Sigma x + x^T b\right) \quad (21)$$

with $\Sigma = \frac{A^T A}{\sigma_y^2}$ and $b = \frac{A^T \sqrt{\alpha_{k_l}}y}{\sigma_y^2}$. Thus, g_{k_l} is proportional to a gaussian distribution $\mathcal{N}(\Sigma^{-1}b, \Sigma^{-1}) = \mathcal{N}(A^{-1}y\sqrt{\alpha_{k_l}}, \sigma_y^2(A^T A)^{-1})$. Now, for all $m \in \{k_l + 1, \dots, k_{l+1}\}$ and $x_m \in \mathbb{R}^d$, we have

$$\begin{aligned}
\hat{g}_m^{l,*}(x_m) &= \int g_{k_l}(x) \hat{p}_{k_l|m}(x|x_m) dx \\
&\propto \int \mathcal{N}(x; A^{-1}y\sqrt{\alpha_{k_l}}, \sigma_y^2(A^T A)^{-1}) \mathcal{N}(x; \mu_{k_l|m}(x_m), \sigma_{k_l|m}^2 I_d) dx \\
&\propto \mathcal{N}(A^{-1}y\sqrt{\alpha_{k_l}}; \mu_{k_l|m}(x_m), \sigma_{k_l|m}^2 I_d + \sigma_y^2(A^T A)^{-1}) \\
&\propto \exp\left(-\frac{1}{2}(A^{-1}y\sqrt{\alpha_{k_l}} - \mu_{k_l|m}(x_m))^T (\sigma_{k_l|m}^2 I_d + \sigma_y^2(A^T A)^{-1})^{-1} (A^{-1}y\sqrt{\alpha_{k_l}} - \mu_{k_l|m}(x_m))\right)
\end{aligned}$$

Where we used lemma 6.2 in the last line. Observing that $A^{-1}y\sqrt{\alpha_{k_l}} - \mu_{k_l|m}(x_m) = A^{-1}(\sqrt{\alpha_{k_l}}y - A\mu_{k_l|m}(x_m))$ we can rewrite the exponent as follows :

$$\begin{aligned}
&(A^{-1}(\sqrt{\alpha_{k_l}}y - \mu_{k_l|m}(x_m)))^T (\sigma_{k_l|m}^2 I_d + \sigma_y^2(A^T A)^{-1})^{-1} (A^{-1}(\sqrt{\alpha_{k_l}}y - \mu_{k_l|m}(x_m))) \\
&= (\sqrt{\alpha_{k_l}}y - A\mu_{k_l|m}(x_m))^T (A(\sigma_{k_l|m}^2 I_d + \sigma_y^2(A^T A)^{-1})A^T)^{-1} (\sqrt{\alpha_{k_l}}y - A\mu_{k_l|m}(x_m)) \\
&= (\sqrt{\alpha_{k_l}}y - A\mu_{k_l|m}(x_m))^T (\sigma_y^2 I + \sigma_{k_l|m}^2 A A^T)^{-1} (\sqrt{\alpha_{k_l}}y - A\mu_{k_l|m}(x_m))
\end{aligned}$$

we can conclude reinjecting this expression into the previous equation. $\square$

Assumption 5.2 (Alternative expression of Loss function expression). *Let $\mu, \nu \in \mathbb{R}^d$. Let $Z, Z' \sim \mathcal{N}(0, I_d)$ and U such that $U|Z \sim \hat{p}_{k_l|m}(\cdot|\mu + e^{\nu/2} \odot Z)$. Then we have the following equality in distribution:*

$$\mathcal{L}_m^l(\mu, \nu, x, Z, U) \sim \tilde{\mathcal{L}}_m^l(\mu, \nu, x, Z, Z')$$

where $\tilde{\mathcal{L}}_m^l(\mu, \nu, x, Z, Z') = \mathcal{L}_m^l(\mu, \nu, x, Z, \mu_{k_l|m}(\mu + e^{\nu/2} \odot Z) + \sigma_{k_l|m} Z')$.

Proof.

$$\begin{aligned} U|Z &\sim \hat{p}_{k_l|m}(\cdot|\mu + e^{\nu/2} \odot Z) \\ &\sim \mathcal{N}(\mu_{k_l|m}(\mu + e^{\nu/2} \odot Z), \sigma_{k_l|m}^2 I_d) \end{aligned}$$

Donc il existe $Z' \sim \mathcal{N}(0, I_d)$ indépendante de Z tel que

$$U|Z = \mu_{k_l|m}(\mu + e^{\nu/2} \odot Z) + \sigma_{k_l|m}^2 Z'$$

Ainsi

$$-\log(g_{k_l}(U))|Z \sim -\log(g_{k_l}(\mu_{k_l|m}(\mu + e^{\nu/2} \odot Z) + \sigma_{k_l|m} Z'))|Z.$$

ce qui conclut la preuve en réinjectant cette égalité dans la définition de $\mathcal{L}_m^l(\mu, \nu, x, Z, U)$. $\square$

Assumption 5.3 (Alternative expression of G). *Let $x \in \mathbb{R}^d$. Let $Z \sim \mathcal{N}(0, I_d)$. Let $Z' \sim p_{k_l|k_{l+1}}(\cdot|x)$ Then for all $x \in \mathbb{R}^d$,*

$$\tilde{G}_\gamma^l(x, Z') \sim \frac{\nabla_x \log g_{k_l}(\mu_{k_l|k_{l+1}}(x) + \sigma_{k_l|k_{l+1}} Z) + \hat{s}_{k_{l+1}}(x)}{\|\nabla_x \log g_{k_l}(\mu_{k_l|k_{l+1}}(x) + \sigma_{k_l|k_{l+1}} Z) + \hat{s}_{k_{l+1}}(x)\|}$$

Proof. Similar as the one of the previous assumption. $\square$

5.2 Gaussian mixtures experiments

Suppose we have data from a gaussian mixture model with C components of equal variance identity, ie

$$p_{data}(x) = \sum_{c=1}^C w_c \mathcal{N}(x; \mu_c, I_d)$$

where for all $c \in \{1, \dots, C\}$, $w_c > 0$, $\sum_{c=1}^C w_c = 1$, $\mu_c \in \mathbb{R}^d$. Suppose we fix $d_y \in \mathbb{N}$, $\sigma_y > 0$, $A \in \mathbb{R}^{d_y \times d}$. Suppose we have a linear inverse problem defined by $y = Ax + \epsilon$ where $\epsilon \sim \mathcal{N}(0, \sigma_y^2 I_{d_y})$. The objective is, given some observation $y \in \mathbb{R}^{d_y}$, to sample from the posterior distribution

$$p(x|y) \propto g_0(x) p_{data}(x)$$

where $g_0(x) = \mathcal{N}(y; Ax, \sigma_y^2 I_{d_y})$.

5.2.1 Explicit expression of the Posterior distribution

The first thing interesting in this particular case is that the posterior distribution has an explicit expression as a gaussian mixture model and thus we can sample from it directly and use these samples to evaluate the performance of the DCPS algorithm.

Assumption 5.4. For all $x \in \mathbb{R}^d$,

$$p(x|y) = \sum_{c=1}^C \tilde{w}_c \mathcal{N}(x; \tilde{\mu}_c, \tilde{\Sigma})$$

where for all $c \in \{1, \dots, C\}$,

$$\begin{aligned}\tilde{\Sigma} &= (I_d + \frac{A^T A}{\sigma_y^2})^{-1} \\ \tilde{\mu}_c &= \tilde{\Sigma}(\mu_c + \frac{A^T y}{\sigma_y^2}) \\ \tilde{w}_c &\propto w_c \mathcal{N}(y; A\mu_c, \sigma_y^2 I_{d_y} + AA^T)\end{aligned}$$

Proof. First, we can rewrite the posterior distribution as follows:

$$\begin{aligned}p(x|y) &\propto g_0(x) p_{data}(x) \\ &= \mathcal{N}(y; Ax, \sigma_y^2 I_{d_y}) \sum_{c=1}^C w_c \mathcal{N}(x; \mu_c, I_d) \\ &= \sum_{c=1}^C w_c \frac{1}{(2\pi)^{(d+d_y)/2} \sigma_y^{d_y}} \exp\left(-\frac{\|y - Ax\|^2}{2\sigma_y^2} - \frac{\|x - \mu_c\|^2}{2}\right)\end{aligned}$$

Now, for all $c \in \{1, \dots, C\}$, we can observe that

$$\begin{aligned}&-\frac{\|y - Ax\|^2}{2\sigma_y^2} - \frac{\|x - \mu_c\|^2}{2} \\ &= -\frac{1}{2}x^T(I_d + \frac{A^T A}{\sigma_y^2})x + x^T(\mu_c + \frac{A^T y}{\sigma_y^2}) + C\end{aligned}$$

where C is a constant independent of x . Thus, we can conclude that

$$\mathcal{N}(y; Ax, \sigma_y^2 I_{d_y}) \mathcal{N}(x; \mu_c, I_d) = m_c(y) \mathcal{N}(x; \tilde{\mu}_c, \tilde{\Sigma}) \quad (22)$$

where $\tilde{\Sigma} = (I_d + \frac{A^T A}{\sigma_y^2})^{-1}$ and $\tilde{\mu}_c = \tilde{\Sigma}(\mu_c + \frac{A^T y}{\sigma_y^2})$ and $m_c(y)$ is a constant independent of x . Reinjecting this expression into the previous equation we get

$$p(x|y) = \sum_{c=1}^C w_c m_c(y) \mathcal{N}(x; \tilde{\mu}_c, \tilde{\Sigma}) \quad (23)$$

ainsi,

$$m_c(y) = \int \mathcal{N}(y; Ax, \sigma_y^2 I_{d_y}) \mathcal{N}(x; \mu_c, I_d) dx$$

This integral can be seen as $\int r(y|x)r(x)dx$ where $r(y|x) = \mathcal{N}(y; Ax, \sigma_y^2 I_{d_y})$ and $r(x) = \mathcal{N}(x; \mu_c, I_d)$ are probability density functions. Thus it can be seen as the marginal of $Y = AX + \epsilon$ where $X \sim \mathcal{N}(\mu_c, I_d)$ and $\epsilon \sim \mathcal{N}(0, \sigma_y^2 I_{d_y})$ independent of X . A linear combinaison of gaussian is gaussian, thus Y is gaussian and we have

$$\begin{aligned}\mathbb{E}[Y] &= A\mu_c \\ \text{Var}(Y) &= A\text{Var}(X)A^T + \text{Var}(\epsilon) = AA^T + \sigma_y^2 I_{d_y}\end{aligned}$$

Thus

$$m_c(y) \propto \mathcal{N}(y; A\mu_c, \sigma_y^2 I_{d_y} + AA^T)$$

which concludes the proof. $\square$

The second interesting point in this particular case is that we can compute explicitly

5.2.2 Explicit expression of the optimal denoising

The second interesting point in this particular case is that we can compute explicitly the optimal denoising prediction $\xi_j^*(x_j)$ for all $j \in \{1, \dots, n\}$ and $x_{0|j}^*(x_j)$. As shown in assumption 2.2,

$$x_{0|j}^*(x_j) = \frac{x_j + (1 - \alpha_j) \nabla_{x_j} \log q_j(x_j)}{\sqrt{\alpha_j}}$$

where q_j is the distribution of $X_j = \sqrt{\alpha_j} X_0 + \sqrt{1 - \alpha_j} Z$ where $Z \sim \mathcal{N}(0, I_d)$. But in the case of gaussian mixture, X_0 is a gaussian mixture, and thus q_j is also a mixture of gaussian z

Assumption 5.5.

$$q_j(x_j) = \sum_{c=1}^C w_c \mathcal{N}(x_j; \sqrt{\alpha_j} \mu_c, I_d)$$

Proof. X_j is a linear combinaison of two independant gaussian. $\square$

Thanks to this result, we have an explicit expression easy to compute for $\nabla \log q_j(x_j)$ because its covariance matrix does not need any inversion and hence an explicit expression for $x_{0|j}^*(x_j)$. As a consequence, we can also compute $\hat{s}_j^*$ very easily. The only other important expression to compute is $\nabla \log g_{k_l}(x)$ because we need it to compute the loss function $\tilde{\mathcal{L}}_m^l$ used in Step 2 and the function $\tilde{G}_\gamma^l$ used in Step 1 of each block of DCPS. But already have shown that $g_{k_l}(x)$ has an explicit expression as a gaussian with covariance matrix proportional to identity, hence $\log g_{k_l}(x)$ is easy to compute because again it needs no matrix inversion.

5.2.3 Implementation details and results

For our experiments on gaussian mixture models, we considered a mixture of 25 gaussians defined as in the paper. The components are $(\mu_c)_{c=1}^{25} = \{(8i, 8j) | i, j \in \{-2, -1, 0, 1, 2\}\}$ and $w_c = 1/25$ for all c . For our implementations, we used the following hyperparameters summarized in Table 1.

We show in Fig 1 four samples obtained with DCPS for different observations. Here are four results of DCPS on gaussian mixture models. We considered a mixture of 25 gaussians defined as in the paper. The components are $(\mu_c)_{c=1}^{25} = \{(8i, 8j) | i, j \in \{-2, -1, 0, 1, 2\}\}$ and $w_c = 1/25$ for all c .

6 Usefull results on gaussians

Table 1: Hyperparameter Settings for DCPS Algorithm

Parameter	Symbol	Value
Data dimension	d_x	10
Observation dimension	d_y	1
Number of GMM components	C	25
Number of blocks	L	3
Total diffusion steps	T	1000
Langevin steps per block	M	50
SGD steps for (μ, v)	K	2
Learning rate for SGD	ζ	0.1
Langevin step size	γ	0.02
Observation noise level	σ_y	0.05

Lemma 6.1 (Tweedie's formula). *Let X be a random variable on $\mathbb{R}^d$ with density p_X . Let Z be a random variable such that $Z|X = x \sim \mathcal{N}(x, \sigma^2 I_d)$ for some $\sigma > 0$. Let p_Z be the density of Z . Then, for almost all $z \in \mathbb{R}^d$, we have*

$$\mathbb{E}[X|Z = z] = z + \sigma^2 \nabla_z \log p_Z(z).$$

Proof. The marginal density of Z is given by

$$p_Z(z) = \int p_X(x) p_{Z|X}(z|x) dx = \int p_X(x) \frac{1}{(2\pi\sigma^2)^{d/2}} \exp\left(-\frac{\|z-x\|^2}{2\sigma^2}\right) dx.$$

Thus $\nabla_z p_Z(z) = \int p_X(x) \frac{-(z-x)}{\sigma^2} p_{Z|X}(z|x) dx$, and we have

$$\begin{aligned} \nabla_z \log p_Z(z) &= \frac{\nabla_z p_Z(z)}{p_Z(z)} \\ &= \frac{1}{p_Z(z)} \int p_X(x) \frac{-(z-x)}{\sigma^2} p_{Z|X}(z|x) dx \\ &= \frac{1}{\sigma^2} \left(\int x \frac{p_X(x) p_{Z|X}(z|x)}{p_Z(z)} dx - z \right) \\ &= \frac{\mathbb{E}[X|Z = z] - z}{\sigma^2}. \end{aligned}$$

□

Lemma 6.2 (Product of two gaussians).

$$\int \mathcal{N}(x; \mu_1, \Sigma_1) \mathcal{N}(x; \mu_2, \Sigma_2) dx = \mathcal{N}(\mu_1; \mu_2, \Sigma_1 + \Sigma_2)$$

Proof. Soit l'intégrale $I = \int_{\mathbb{R}^d} \mathcal{N}(x; \mu_1, \Sigma_1) \mathcal{N}(x; \mu_2, \Sigma_2) dx$. Par définition des densités gaussiennes :

$$I = \int_{\mathbb{R}^d} \frac{1}{(2\pi)^d \sqrt{|\Sigma_1| |\Sigma_2|}} \exp\left(-\frac{1}{2} [\|x - \mu_1\|_{\Sigma_1}^2 + \|x - \mu_2\|_{\Sigma_2}^2]\right) dx$$

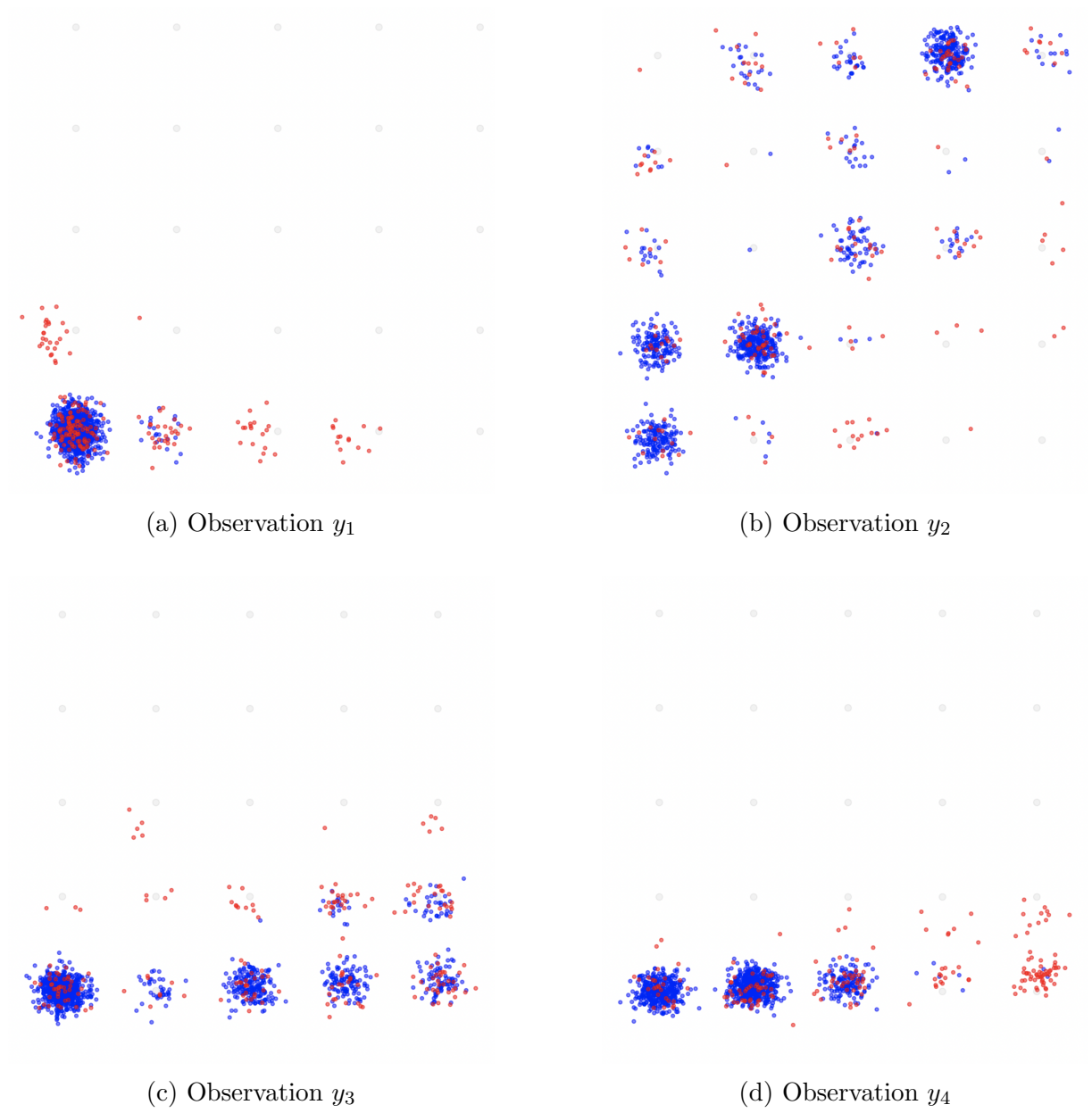


Figure 1: Samples generated by DCPS for four different observations in a GMM setting.

Posons

$$\begin{aligned}
 Q(x) &= (x - \mu_1)^\top \Sigma_1^{-1} (x - \mu_1) + (x - \mu_2)^\top \Sigma_2^{-1} (x - \mu_2) \\
 &= x^\top (\Sigma_1^{-1} + \Sigma_2^{-1}) x - 2x^\top (\Sigma_1^{-1} \mu_1 + \Sigma_2^{-1} \mu_2) + \mu_1^\top \Sigma_1^{-1} \mu_1 + \mu_2^\top \Sigma_2^{-1} \mu_2
 \end{aligned}$$

Posons $\Sigma_c^{-1} = \Sigma_1^{-1} + \Sigma_2^{-1}$ et $\mu_c = \Sigma_c (\Sigma_1^{-1} \mu_1 + \Sigma_2^{-1} \mu_2)$. En complétant le carré en x , on obtient :

$$Q(x) = (x - \mu_c)^\top \Sigma_c^{-1} (x - \mu_c) + K$$

Le terme constant K est donné par :

$$\begin{aligned}
K &= \mu_1^\top \Sigma_1^{-1} \mu_1 + \mu_2^\top \Sigma_2^{-1} \mu_2 - \mu_c^\top \Sigma_c^{-1} \mu_c \\
&= \mu_1^\top \Sigma_1^{-1} \mu_1 + \mu_2^\top \Sigma_2^{-1} \mu_2 - (\Sigma_1^{-1} \mu_1 + \Sigma_2^{-1} \mu_2)^\top \Sigma_c (\Sigma_1^{-1} \mu_1 + \Sigma_2^{-1} \mu_2) \\
&= \mu_1^\top \Sigma_1^{-1} \mu_1 + \mu_2^\top \Sigma_2^{-1} \mu_2 - \mu_1^\top \Sigma_1^{-1} \Sigma_c \Sigma_1^{-1} \mu_1 - \mu_2^\top \Sigma_2^{-1} \Sigma_c \Sigma_2^{-1} \mu_2 - 2\mu_1^\top \Sigma_1^{-1} \Sigma_c \Sigma_2^{-1} \mu_2 \\
&= \mu_1^\top (\Sigma_1^{-1} - \Sigma_1^{-1} \Sigma_c \Sigma_1^{-1}) \mu_1 + \mu_2^\top (\Sigma_2^{-1} - \Sigma_2^{-1} \Sigma_c \Sigma_2^{-1}) \mu_2 - 2\mu_1^\top \Sigma_1^{-1} \Sigma_c \Sigma_2^{-1} \mu_2
\end{aligned}$$

Or

$$\begin{aligned}
\Sigma_1^{-1} - \Sigma_1^{-1}(\Sigma_1^{-1} + \Sigma_2^{-1})^{-1}\Sigma_1^{-1} &= (\Sigma_1 + \Sigma_2)^{-1} \\
\Sigma_2^{-1} - \Sigma_2^{-1}(\Sigma_1^{-1} + \Sigma_2^{-1})^{-1}\Sigma_2^{-1} &= (\Sigma_1 + \Sigma_2)^{-1} \\
-\Sigma_1^{-1}(\Sigma_1^{-1} + \Sigma_2^{-1})^{-1}\Sigma_2^{-1} &= -(\Sigma_1 + \Sigma_2)^{-1}
\end{aligned}$$

Donc

$$K = (\mu_1 - \mu_2)^\top (\Sigma_1 + \Sigma_2)^{-1} (\mu_1 - \mu_2)$$

L'intégrale devient alors :

$$\begin{aligned}
I &= \frac{e^{-K/2}}{(2\pi)^d \sqrt{|\Sigma_1| |\Sigma_2|}} \int_{\mathbb{R}^d} \exp \left(-\frac{1}{2} (x - \mu_c)^\top \Sigma_c^{-1} (x - \mu_c) \right) dx \\
&= \frac{e^{-K/2}}{(2\pi)^d \sqrt{|\Sigma_1| |\Sigma_2|}} \cdot (2\pi)^{d/2} \sqrt{|\Sigma_c|} \\
&= \frac{1}{(2\pi)^{d/2} \sqrt{\frac{|\Sigma_1| |\Sigma_2|}{|\Sigma_c|}}} \exp \left(-\frac{1}{2} (\mu_1 - \mu_2)^\top (\Sigma_1 + \Sigma_2)^{-1} (\mu_1 - \mu_2) \right)
\end{aligned}$$

En utilisant la propriété du déterminant $|\Sigma_1| |\Sigma_2| |\Sigma_c^{-1}| = |\Sigma_1| |\Sigma_2| |\Sigma_1^{-1} + \Sigma_2^{-1}| = |\Sigma_1 + \Sigma_2|$, nous obtenons enfin :

$$I = \frac{1}{(2\pi)^{d/2} \sqrt{|\Sigma_1 + \Sigma_2|}} \exp \left(-\frac{1}{2} (\mu_1 - \mu_2)^\top (\Sigma_1 + \Sigma_2)^{-1} (\mu_1 - \mu_2) \right).$$

□

Lemma 6.3 (Distance between coupling of gaussians). *Let $\epsilon \sim \mathcal{N}(0, I_d)$, $\mu_1, \mu_2 \in \mathbb{R}^d$ and $\sigma_1, \sigma_2 \in \mathbb{R}_+^*$. Let X and Y be random variables such that*

$$X = \mu_1 + \sigma_1 \epsilon, \quad Y = \mu_2 + \sigma_2 \epsilon.$$

Then, $\mathbb{E} [\|X - Y\|^2] = \|\mu_1 - \mu_2\|^2 + d(\sigma_1 - \sigma_2)^2$.

Proof. We have

$$\begin{aligned}
\mathbb{E} [\|X - Y\|^2] &= \mathbb{E} [\|(\mu_1 - \mu_2) + (\sigma_1 - \sigma_2)\epsilon\|^2] \\
&= \mathbb{E} [\|\mu_1 - \mu_2\|^2 + 2(\mu_1 - \mu_2)^\top (\sigma_1 - \sigma_2)\epsilon + (\sigma_1 - \sigma_2)^2 \|\epsilon\|^2] \\
&= \|\mu_1 - \mu_2\|^2 + 0 + (\sigma_1 - \sigma_2)^2 \mathbb{E} [\|\epsilon\|^2] \\
&= \|\mu_1 - \mu_2\|^2 + d(\sigma_1 - \sigma_2)^2.
\end{aligned}$$

□

Lemma 6.4 (KL divergence between gaussian distributions). *Let $\mu_1, \mu_2 \in \mathbb{R}^d$ and Σ_1, Σ_2 be two positive definite matrices in $\mathbb{R}^{d \times d}$. Let $P_1 = \mathcal{N}(\mu_1, \Sigma_1)$ and $P_2 = \mathcal{N}(\mu_2, \Sigma_2)$ be two Gaussian distributions. Then,*

$$KL(P_1||P_2) = \frac{1}{2} \left[tr(\Sigma_2^{-1}\Sigma_1) + (\mu_2 - \mu_1)^\top \Sigma_2^{-1}(\mu_2 - \mu_1) - d + \log \left(\frac{|\Sigma_2|}{|\Sigma_1|} \right) \right]. \quad (24)$$

Proof. Straightforward calculations. □